



EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E WEB PROGRAMMING AND INTERFACE DESIGN

OVERVIEW

- **What** is JQuery?
- **Why** learn JQuery?
- JQuery features/**syntax**
- Examples
- Questions and Discussion

REVISION QUESTION

What is the result?

```
(true + false) > 2 + true
```

- A. true
- B. false
- C. TypeError
- D. NaN

ANSWER

Answer: **B.** false

$(\text{true} + \text{false}) = 1, 2 + \text{true} = 3$

REVISION QUESTION

What is the result?

```
new Array(5).toString();
```

- A. []
- B. "[]"
- C. ".,.,."

ANSWER

Answer: C. ".,,"

REVISION QUESTION

What is the output?

```
var foo = function foo(){  
  alert( foo==foo );  
};  
foo();
```

- A. true
- B. false
- C. ReferenceError

ANSWER

Answer: **A.** true

WHAT IS JQUERY?



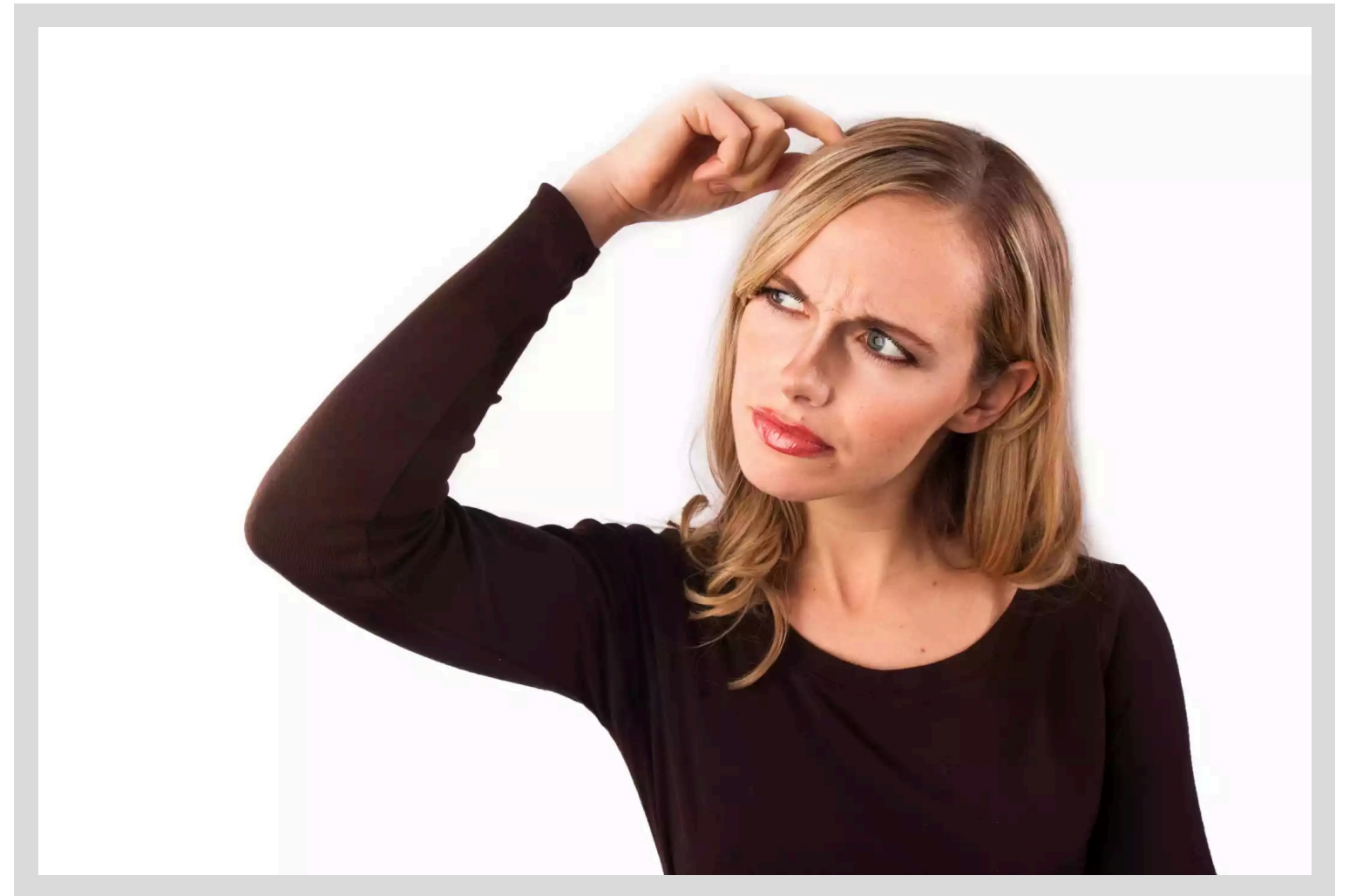
WHAT IS JQUERY?

- jQuery is a **JavaScript Library**
- jQuery greatly **simplifies** JavaScript programming
- jQuery is easy to learn (uses CSS syntax for selection)
- **Popular** - most used Javascript library on the web today
 - 30% of all sites with Javascript use jQuery * *
 - According to Stack Overflow's **2023** survey, 22.87% of developers still use JQuery, it's the third most popular web framework and one of the most starred repositories on GitHub *
- **Cross browser** compatibility (IE6+, FF2+, ..)

FEATURES OF JQUERY

- **DOM** element selection
- DOM traversal and modification (including support for CSS 1-3)
- DOM manipulation based on CSS selectors that uses node element names and node element attributes (id and class)
- Events
- **Effects and animations**
- jQuery provides several methods for AJAX functionality
- Extensibility (through plugins)
- Cross browser support

WHO USES JQUERY?



WHO USES JQUERY?

Google, Amazon, **IBM**, Twitter, NBC, BBC, Digg, **Intel** Oracle, Slashdot, Technoratis, Sourceforge, GitHub, New York Post, Feedburner, ...

Mozilla, Wordpress, Drupal, Joomla, Symfony, Zend, Framework, PEAR, ...

PROS AND CONS OF JQUERY

Pros:

- JQuery is easy to use and understand
- The core library is compact and small (<15kb)
- Their statement/strategy is: **Write Less, Do More, Javascript Library**

Cons:

- Hard to use with OOP
- JQuery supports plugins (but all these plugins are not **verified**)
- JQuery is a **library**

JQUERY - NOT THE ONLY TOOL IN THE BAG

- Dojo Toolkit
- YUI (Yahoo User Interface Library)
- MooTools
- ExtJS

However, JQuery is the most **popular**

JQUERY (MOTTO)

Taken from the JQuery website:

*jQuery is a **fast** and concise JavaScript Library that **simplifies** HTML document traversing, event handling, animating, and Ajax interactions for **rapid web development**. jQuery is designed to change the way that you write JavaScript*

DOM AND JQUERY

- Browser needs to load jQuery (**library**)
- Uses the DOM

QUESTION

How can you be sure your (JavaScript/jQuery) code/library is **ready** to run?

Answer: Wrap all your jQuery code within the document **ready function**

```
$(document).ready(function() {  
    // insert juicy jQuery code here ...  
});
```

See [link](#) for details

WHAT HAPPENS IF WE **DON'T** CHECK IF THINGS ARE
READY?



WHAT HAPPENS IF WE DON'T CHECK IF THINGS ARE READY?

1. Code doesn't work (90% of time might throw an error)
2. Code works .. the page loads, next page loads, then fails (**unreliable**)

JQUERY CONCEPTS

MAGIC **\$()** FUNCTION

MAGIC **\$()** FUNCTION

```
var el = $( "<div/>" );
```

- Create HTML elements on-the-fly (dynamically)

MAGIC **\$()** FUNCTION

```
$(window).width();
```

- Manipulate existing DOM elements

MAGIC **\$()** FUNCTION

```
$( "div" ).hide();  
$( "div, p" ).hide();
```

- Select document elements

See [link](#) for details

MAGIC **\$()** FUNCTION

```
$(document).ready(function(){..});
```

- Called when the document is **ready** for programming

LOADING JQUERY

- To use jQuery you need to load jQuery **library**
 - [\(jQuery Homepage\)](#)
- You can include it locally on your own server

```
<script src="/js/jquery.js">
```

- Or use one of the available resource links:

```
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js"></script>
```

```
<script src="ajax.microsoft.com/ajax/jquery/jquery-3.4.1.js"></script>
```



EXAMPLES JQUERY

Download/API details/tutorials/examples.

[jQuery.com](http://jquery.com)

LOAD SCRIPTS AT THE BOTTOM

Why?

- When scripts are downloading they block everything else in most browsers
- Good practice to load your scripts at the bottom of your page so they don't interrupt the page content downloads
- The **order** scripts are loaded and used matters

JQUERY **SYNTAX**

- jQuery syntax is tailer **made** for **selecting** HTML elements and performing actions on the element(s)
- **Friendly** easy to use syntax
- Basic syntax is

```
$(selector).action()
```

- A **\$** dollar sign to define jQuery
- A **(selector)** to "query (or find)" HTML elements
- A jQuery **action()** to be performed on the element(s)

ALL SELECTOR

```
$("*") // find everything
```

- Selectors return pseudo-array of jQuery elements

BASIC SELECTORS

- by Tag

```
$( "div" ) // <div>Hello jQuery</div>
```

- by ID

```
$( "#usr" ) // <span id="usr">John</span>
```

- by Class

```
$( ".menu" ) // <ul class="menu">Home</ul>
```

SIMPLE JQUERY EXAMPLE

```
<html><head>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js">
</script>
<script>
$(document).ready(function(){
    $("p").click(function(){
        $(this).hide();
    });
});
</script>
</head><body>
<p>If you click on me, I will disappear.</p>
<p>Click anywhere</p>
</body></html>
```

REVIEW

- HTML:

```
<p>Hello</p>
```

- Script

```
$(function(){  
  $("p").addClass("powerful");  
});
```

- Result:

```
<p class="powerful">Hello</p>
```

REVIEW

Note

```
$(function(){ .... } );
```

is shorthand for

```
$(document).ready(function(){ ... } );
```

SELECTORS

Basic Selectors

- `$("p")`

```
<p>
```

- `$("div")`

```
<div>
```

- `$("#foo")`

```
<p id="foo"> ..
```

- `$(".foo")`

```
<p class="foo"> ..
```

Further details see api.jquery.com

EXAMPLE

- jQuery

```
$( "p" ).removeClass( "cats" );
```

- Before:

```
<p class="cats">
```

- After

```
<p class="">
```

EXAMPLE "<div>" HIDE/SHOW

- jQuery

```
$( "div" ).show( );
```

- Before:

```
<div style="display:none;">
```

- After

```
<div style="display:block;">
```

EXAMPLE IDS

- jQuery

```
$( "#bob" ).text( "Blue" );
```

- Before:

```
<p id="bob">What colour?</p>
```

- After

```
<p id="bob">Blue</p>
```

CHAINING

Chain most methods together:

```
$( "p" )  
  .addClass( "cat" )  
  .text( "Hello" )  
  .show();
```

COMMON METHODS

- **.show()** - show hidden element
- **.wrap("<a>")** - wrap an element with <a>
- **.parent("p")** - select parent <p>
- **.html()** - get/set innerHTML
- **.val()** - get/set value

Selectors

```
$( "*") // all elements
$( "p.demo") // <p> elements with class="demo"
$( "p:first") // the first <p> element
$( "p span") // span, descendant of p
$( "p > span") // span, direct child of p
$( "p + span") // span immediately proceeded by a p
$( "p ~ span") // strong element proceeded by p
$( "ul li:first") // the first <li> element of the first <ul>
$( "ul li:first-child") // the first <li> element of every <ul>
$( "ul li:nth-child(3)") // third child
$( "[href]") // any element with an href attribute
$( "a[target='_blank']") // <a> elements with a target "_blank" attribute
$( "a[target!='_blank']") // <a> elements with a target attribute value other than "_blank"
$( ":input") // all form elements
$( ":button") // <button> and <input> elements of type="button"
$( "tr:even") // even <tr> elements
$( "tr:odd") // odd <tr> elements
$( "span:parent") // element which has child element
$( "span:contains('demo')") // element conaining the specified text
```

Actions

```
$(selector).action()
$(this).hide() // the current element
$( "div").hide() // all <div> elements
$( ".demo").hide() // all elements with class="demo"
$( "#demo").hide() // the element with id="demo"
```

Events

```
$( ".demo" ).click(function(){
$(this).hide(200);
});
```

Mouse

scroll, click, dblclick, mousedown, mouseup, mousemove, mouseover, mouseout, mouseenter, mouseleave, error

Keyboard

keydown, keypress, keyup

Form

submit, change, focus, blur

Basics

Include

[Download](#)

```
<script src="https://code.jquery.com/jquery-latest.min.js" type="text/javascript"></script>
<script src="https://code.jquery.com/jquery-3.2.0.min.js"></script>
```

```
// parse each .demo element
output their text
```

bind an event on an element

avoid conflict with other frameworks also using the dollar sign

});

Ads

CHEAT SHEET

Easily lookup jQuery features

Online resources

<https://htmlcheatsheet.com/jquery/>

PLUGINS

- [Plugins.jquery.com](https://plugins.jquery.com)
- Plugins offer powerful features to add animations, effects or UI components

QUESTION

What is the correct selector to hide all <p> elements?

```
$( "___" ).hide();
```

ANSWER

What is the correct selector to hide all <p> elements?

```
$("p").hide();
```

QUESTION

What is the jQuery method to return the text content of a `<div>` element?

```
$("div").text();
```

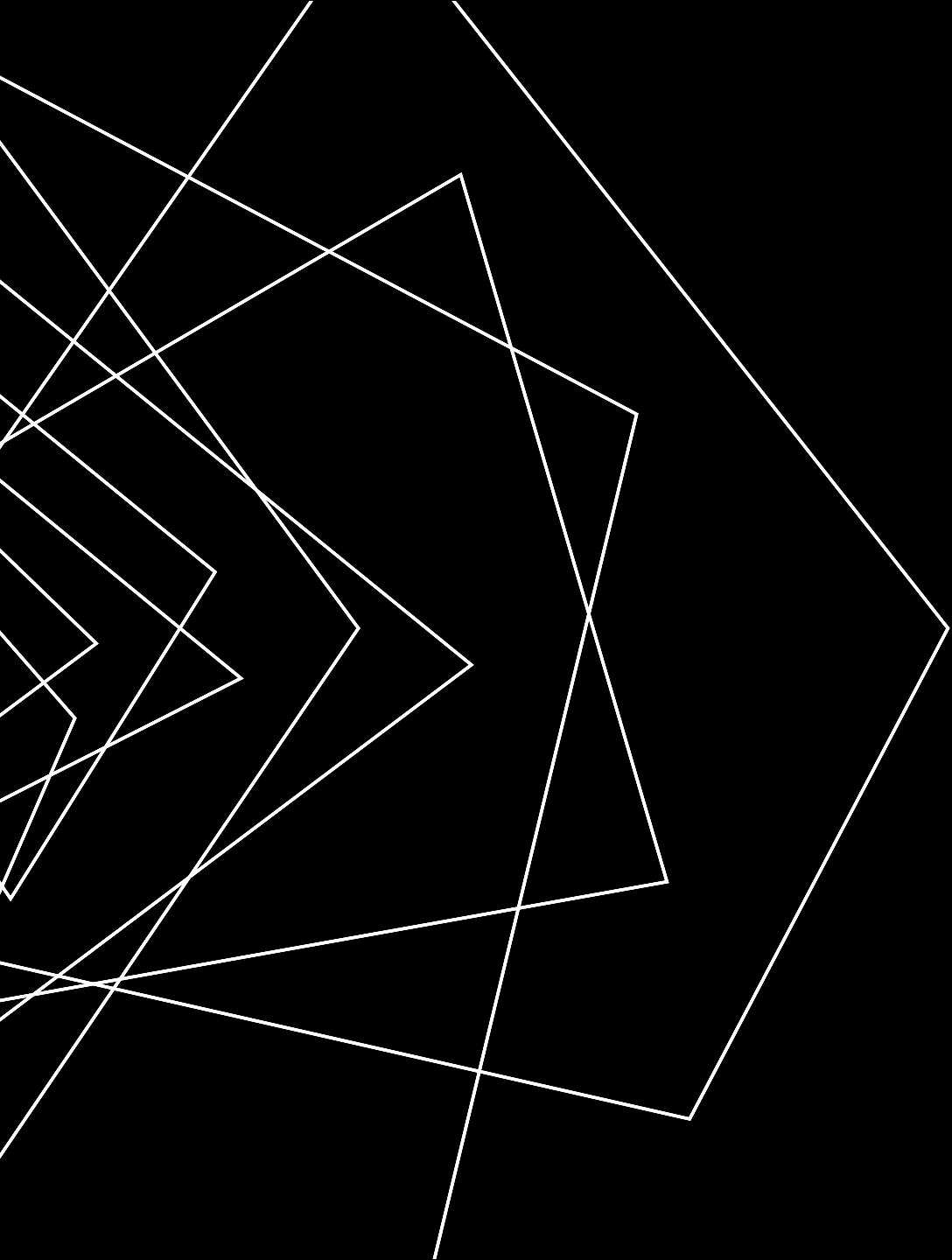
ANSWER

What is the jQuery method to return the text content of a `<div>` element?

```
$("div").text();
```

SUMMARY

- Understand **JQuery** and how to use it in web development
- Essential **syntax** and methods
- Basic examples



THANK YOU

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